

Day 3

R Basic Exercises

Using Numeric Variables

Basic

```
n1 <- 2
n2 <- 18
x <- 243
sum_product <- 213457

some_sum <- sum(n1, n2)
some_length <- x + 8.3
sum_product <- sum_product * x
```

```
x <- 2
y <- 3
z <- 4

three_x <- 3 * x
three_x_plus_y <- three_x + y
x_plus_y_div_seven <- (x + y) / 7
three_x_plus_seven_div_z_plus_two <- (three_x + 7) / (z + 2)
```

Intermediate

```
n1 <- 10
n2 <- 2

x <- n1 %/% n2
y <- n1 %% n2

print(x)
```

```
[1] 5
```

```
print(y)
```

```
[1] 0
```

```
# Original Code
# f <- 98.6
# c <- 5/9 * f-32
# print(c)

#Working code
f <- 98.6
c <- (f-32) * (5/9) # Make it follow PEMDAS
print(c)
```

```
[1] 37
```

```
ton <- 35273.92
fruity_pebbles <- 12

in_tons <- (fruity_pebbles/ton)
print(in_tons)
```

```
[1] 0.0003401947
```

```
how_many_boxes <- (ton/fruity_pebbles)
print(how_many_boxes)
```

```
[1] 2939.493
```

Advanced

```
total_pay <- 100
salary <- 0
contribute <- 0

salary <- (total_pay * 0.05)

take_home <- total_pay - salary

mpf <- total_pay - take_home

contribute <- mpf

print(take_home)
```

[1] 95

```
print(mpf)
```

[1] 5

```
print(contribute)
```

[1] 5

```
total_cost <- take_home + mpf + contribute
print(total_cost)
```

[1] 105

```
total_pay <- 105
salary <- total_cost / 1.05
mpf <- salary * 0.05
take_home <- salary - mpf
print(salary)
```

[1] 100

```
print(mpf)
```

```
[1] 5
```

```
print(take_home)
```

```
[1] 95
```

Using Text String Variables

Basic + Intermediate

```
word1 <- "ointment"  
word2 <- "perchance"  
word3 <- "unguard"  
paste0(word1, word2, word3)
```

```
[1] "ointmentperchanceunguard"
```

```
paste(word1, word2, word3, sep=", ")
```

```
[1] "ointment, perchance, unguard"
```

```
substr(word1, 3, 3)
```

```
[1] "n"
```

```
substr(word2, 3, 3)
```

```
[1] "r"
```

```
substr(word3, 3, 3)
```

```
[1] "g"
```

```
word1_data <- nchar(word1)
word2_data <- nchar(word2)
word3_data <- nchar(word3)

substr(word1, word1_data, word1_data)
```

```
[1] "t"
```

```
substr(word2, word1_data, word1_data)
```

```
[1] "c"
```

```
substr(word3, word3_data, word3_data)
```

```
[1] "d"
```

```
substr(word1, nchar(word1) - 2, nchar(word1))
```

```
[1] "ent"
```

```
substr(word2, nchar(word2) - 2, nchar(word2))
```

```
[1] "nce"
```

```
substr(word3, nchar(word3) - 2, nchar(word3))
```

```
[1] "ard"
```

Advanced

```
sentence <- "Hello, I am your Overload."
words <- strsplit(sentence, " ")
print(words)
```

```
[[1]]
[1] "Hello,"      "I"           "am"          "your"        "Overload."
```

```
Instructor <- "Steve"
name <- "Braden"
food <- "Skittles"
number <- 111
adjective <- "floppy"
color <- "pink"
animal <- "dragon"
print(paste("Dear Instructor", Instructor, ",", "I am sorry that I am unable to turn in my h
```

```
[1] "Dear Instructor Steve , I am sorry that I am unable to turn in my homework at this time
```

Using Data Frames

Basic + Intermediate

```
library(languageR)
data <- lexdec
print(data$Frequency)
```

```
[1] 4.859812 4.605170 4.997212 4.727388 7.667626 4.060443 4.753590 4.709530
[9] 3.044522 4.204693 5.298317 4.043051 2.484907 5.631212 5.700444 6.228511
[17] 1.791759 5.545177 4.595120 5.918894 4.442651 2.995732 3.044522 5.267858
[25] 3.970292 4.976734 4.615121 4.624973 4.844187 5.192957 5.652489 6.461468
[33] 5.129899 3.555348 3.663562 5.805135 4.007333 3.988984 4.812184 6.577861
[41] 4.454347 2.079442 4.418841 6.660575 6.109248 5.880533 4.248495 6.120297
[49] 5.541264 2.833213 3.828641 3.332205 3.663562 5.023881 6.098074 6.063785
[57] 3.637586 5.117994 6.304449 3.044522 4.682131 7.086738 2.484907 5.347108
[65] 3.332205 3.951244 7.771910 5.537334 2.890372 2.708050 6.378426 3.433987
[73] 6.599870 5.214936 5.587249 4.962845 4.499810 4.127134 5.783825 7.667626
[81] 3.828641 3.951244 4.007333 4.499810 6.109248 5.587249 5.298317 5.541264
[89] 4.812184 4.418841 6.228511 3.637586 4.844187 3.970292 4.442651 3.044522
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1	A1	6.340359	23	F	English	correct	word	correct
2	A1	6.308098	27	F	English	correct	nonword	correct
3	A1	6.349139	29	F	English	correct	nonword	correct
4	A1	6.186209	30	F	English	correct	word	correct
5	A1	6.025866	32	F	English	correct	nonword	correct
6	A1	6.180017	33	F	English	correct	word	correct
7	A1	6.035481	34	F	English	correct	word	correct
8	A1	6.263398	38	F	English	correct	nonword	correct
9	A1	6.459904	41	F	English	correct	nonword	correct
10	A1	6.184149	42	F	English	correct	word	correct
11	A1	6.137727	45	F	English	correct	nonword	correct
12	A1	6.242223	46	F	English	correct	word	correct
13	A1	6.003887	47	F	English	correct	word	correct
14	A1	6.035481	48	F	English	correct	word	correct
15	A1	6.322565	53	F	English	correct	nonword	correct
16	A1	5.988961	54	F	English	correct	word	correct
17	A1	6.285998	55	F	English	correct	word	correct
18	A1	6.381816	58	F	English	correct	nonword	correct
19	A1	5.958425	59	F	English	correct	word	correct
20	A1	6.124683	61	F	English	incorrect	nonword	correct
21	A1	6.741701	62	F	English	correct	word	incorrect

22	A1	6.821107	64	F	English	correct	nonword	correct
23	A1	6.406880	65	F	English	correct	word	correct
24	A1	6.421622	67	F	English	correct	nonword	correct
25	A1	6.320768	68	F	English	correct	word	correct
26	A1	6.208590	75	F	English	correct	nonword	correct
27	A1	6.214608	76	F	English	correct	word	correct
28	A1	7.114769	79	F	English	correct	nonword	correct
29	A1	6.091310	80	F	English	correct	word	correct
30	A1	6.208590	82	F	English	correct	nonword	correct
31	A1	6.366470	88	F	English	correct	nonword	correct
32	A1	6.582025	91	F	English	correct	nonword	correct
33	A1	6.514713	93	F	English	correct	nonword	correct
34	A1	6.352629	94	F	English	correct	word	correct
35	A1	6.278521	96	F	English	correct	nonword	correct
36	A1	5.988961	97	F	English	correct	word	correct
37	A1	6.079933	98	F	English	correct	word	correct
38	A1	6.126869	100	F	English	correct	nonword	correct
39	A1	6.284134	102	F	English	correct	nonword	correct
40	A1	6.089045	106	F	English	correct	nonword	correct
41	A1	6.383507	108	F	English	correct	nonword	correct
42	A1	6.220590	109	F	English	correct	word	correct
43	A1	6.381816	111	F	English	correct	nonword	correct
44	A1	6.202536	113	F	English	correct	nonword	correct
45	A1	6.356108	119	F	English	correct	nonword	correct
46	A1	6.107023	120	F	English	correct	word	correct
47	A1	6.517671	123	F	English	correct	nonword	correct
48	A1	6.317165	125	F	English	correct	nonword	correct
49	A1	6.086775	126	F	English	correct	word	correct
50	A1	6.163315	127	F	English	correct	word	correct
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53	A1	6.333280	134	F	English	correct	word	correct
54	A1	6.202536	135	F	English	correct	word	correct
55	A1	6.111467	136	F	English	correct	word	correct
56	A1	6.326149	139	F	English	correct	nonword	correct
57	A1	6.903747	141	F	English	correct	nonword	correct
58	A1	6.306275	144	F	English	correct	nonword	correct
59	A1	6.118097	146	F	English	correct	nonword	correct
60	A1	6.472346	148	F	English	correct	nonword	correct
61	A1	6.697034	151	F	English	correct	nonword	correct
62	A1	6.148468	152	F	English	correct	word	correct
63	A1	6.599870	153	F	English	correct	word	correct
64	A1	6.775366	157	F	English	correct	nonword	correct

65	A1	6.366470	162	F	English	correct	nonword	correct
66	A1	6.274762	163	F	English	correct	word	correct
67	A1	6.023448	164	F	English	correct	word	correct
68	A1	6.139885	167	F	English	correct	nonword	incorrect
69	A1	6.293419	168	F	English	correct	word	correct
70	A1	6.021023	169	F	English	correct	word	correct
71	A1	6.188264	170	F	English	correct	word	correct
72	A1	6.194405	173	F	English	correct	nonword	correct
73	A1	5.899897	174	F	English	correct	word	correct
74	A1	6.364751	177	F	English	correct	nonword	correct
75	A1	5.973810	179	F	English	correct	nonword	correct
76	A1	6.037871	180	F	English	correct	word	correct
77	A1	5.973810	182	F	English	correct	nonword	correct
78	A1	6.282267	184	F	English	correct	nonword	correct
79	A1	6.040255	185	F	English	correct	word	correct

		Word	Frequency	FamilySize	SynsetCount	Length	Class	FreqSingular
1		owl	4.859812	1.3862944	0.6931472	3	animal	54
2		mole	4.605170	1.0986123	1.9459101	4	animal	69
3		cherry	4.997212	0.6931472	1.6094379	6	plant	83
4		pear	4.727388	0.0000000	1.0986123	4	plant	44
5		dog	7.667626	3.1354942	2.0794415	3	animal	1233
6		blackberry	4.060443	0.6931472	1.3862944	10	plant	26
7		strawberry	4.753590	0.0000000	1.3862944	10	plant	50
8		squirrel	4.709530	0.0000000	1.0986123	8	animal	63
9		radish	3.044522	0.6931472	1.3862944	6	plant	11
10		almond	4.204693	0.0000000	1.0986123	6	plant	24
11		whale	5.298317	1.0986123	1.3862944	5	animal	104
12		reindeer	4.043051	0.0000000	0.6931472	8	animal	28
13		blueberry	2.484907	0.0000000	1.0986123	9	plant	7
14		lemon	5.631212	0.6931472	1.6094379	5	plant	233
15		bee	5.700444	1.7917595	1.0986123	3	animal	118
16		goat	6.228511	1.9459101	1.0986123	4	animal	209
17		eggplant	1.791759	0.0000000	1.0986123	8	plant	4
18		tomato	5.545177	0.0000000	1.0986123	6	plant	123
19		peanut	4.595120	0.6931472	1.6094379	6	plant	49
20		bat	5.918894	2.1972246	2.3025851	3	animal	155
21		mustard	4.442651	0.0000000	1.3862944	7	plant	84
22		magpie	2.995732	0.0000000	1.0986123	6	animal	13
23		stork	3.044522	0.0000000	0.6931472	5	animal	13
24		goose	5.267858	1.7917595	1.3862944	5	animal	103
25		squid	3.970292	0.0000000	1.0986123	5	animal	23
26		carrot	4.976734	0.6931472	1.6094379	6	plant	45
27		crocodile	4.615121	0.0000000	0.6931472	9	animal	35

28	tortoise	4.624973	0.0000000	0.6931472	8 animal	71
29	spider	4.844187	0.6931472	1.0986123	6 animal	73
30	grape	5.192957	1.0986123	1.0986123	5 plant	37
31	fox	5.652489	1.7917595	1.9459101	3 animal	177
32	potato	6.461468	0.0000000	1.0986123	6 plant	206
33	frog	5.129899	2.0794415	1.3862944	4 animal	74
34	beetroot	3.555348	0.0000000	1.0986123	8 plant	15
35	asparagus	3.663562	0.0000000	1.0986123	9 plant	19
36	mouse	5.805135	1.9459101	1.6094379	5 animal	146
37	dolphin	4.007333	0.0000000	1.0986123	7 animal	24
38	pineapple	3.988984	0.0000000	1.0986123	9 plant	44
39	lettuce	4.812184	0.0000000	1.0986123	7 plant	115
40	sheep	6.577861	2.3025851	1.3862944	5 animal	359
41	cucumber	4.454347	0.0000000	1.0986123	8 plant	54
42	gherkin	2.079442	0.0000000	1.0986123	7 plant	4
43	gull	4.418841	1.6094379	1.0986123	4 animal	23
44	pig	6.660575	2.7725887	2.3025851	3 animal	320
45	camel	6.109248	0.6931472	0.6931472	5 animal	146
46	shark	5.880533	0.6931472	1.7917595	5 animal	246
47	vulture	4.248495	0.0000000	1.0986123	7 animal	21
48	snake	6.120297	1.6094379	1.9459101	5 animal	251
49	donkey	5.541264	0.0000000	1.0986123	6 animal	162
50	broccoli	2.833213	0.0000000	1.0986123	8 plant	16
51	apricot	3.828641	0.0000000	1.3862944	7 plant	26
52	leek	3.332205	0.0000000	1.0986123	4 plant	5
53	clove	3.663562	0.0000000	1.6094379	5 plant	16
54	banana	5.023881	0.0000000	1.0986123	6 plant	74
55	lion	6.098074	1.0986123	1.0986123	4 animal	152
56	elephant	6.063785	0.6931472	1.0986123	8 animal	113
57	hedgehog	3.637586	0.0000000	1.0986123	8 animal	21
58	eagle	5.117994	1.0986123	1.7917595	5 animal	129
59	apple	6.304449	1.0986123	1.0986123	5 plant	315
60	kiwi	3.044522	0.0000000	1.3862944	4 plant	11
61	wasp	4.682131	1.3862944	0.6931472	4 animal	43
62	cat	7.086738	2.7725887	2.1972246	3 animal	739
63	paprika	2.484907	0.0000000	1.0986123	7 plant	11
64	ant	5.347108	1.3862944	1.0986123	3 animal	69
65	bunny	3.332205	0.0000000	1.0986123	5 animal	19
66	beaver	3.951244	0.0000000	2.0794415	6 animal	37
67	horse	7.771910	3.3322045	2.0794415	5 animal	1518
68	mushroom	5.537334	0.6931472	1.9459101	8 plant	90
69	woodpecker	2.890372	0.0000000	0.6931472	10 animal	11
70	moose	2.708050	0.0000000	0.6931472	5 animal	7

71	orange	6.378426	1.3862944	1.7917595	6	plant	258
72	avocado	3.433987	0.0000000	1.3862944	7	plant	24
73	chicken	6.599870	1.0986123	1.7917595	7	animal	534
74	butterfly	5.214936	0.0000000	1.0986123	9	animal	88
75	olive	5.587249	1.0986123	1.9459101	5	plant	160
76	swan	4.962845	1.3862944	1.3862944	4	animal	88
77	walnut	4.499810	0.0000000	1.3862944	6	plant	70
78	melon	4.127134	1.0986123	1.0986123	5	plant	38
79	monkey	5.783825	0.0000000	1.6094379	6	animal	162

	FreqPlural	DerivEntropy	Complex	rInfl	meanRT	SubjFreq	meanSize
1	74	0.7912	simplex	-0.31015493	6.3582	3.12	3.4758
2	30	0.6968	simplex	0.81450804	6.4150	2.40	2.9999
3	49	0.4754	simplex	0.51879379	6.3426	3.88	1.6278
4	68	0.0000	simplex	-0.42744401	6.3353	4.52	1.9908
5	828	1.2129	simplex	0.39779605	6.2956	6.04	4.6429
6	31	0.3492	complex	-0.16989904	6.3959	3.28	1.5831
7	65	0.0000	complex	-0.25782911	6.3475	5.04	1.8922
8	47	0.0000	simplex	0.28768207	6.4491	2.80	3.7114
9	9	0.9403	simplex	0.18232156	6.4133	3.12	1.3226
10	42	0.0000	simplex	-0.54232429	6.3908	3.72	1.8912
11	95	0.4994	simplex	0.08961216	6.3548	3.96	3.5350
12	28	0.0000	complex	0.00000000	6.4974	2.28	3.7494
13	4	0.0000	complex	0.47000363	6.3549	3.76	2.0858
14	45	0.6285	simplex	1.62667972	6.2900	5.04	1.8119
15	180	0.8521	simplex	-0.41937354	6.2978	4.56	3.6399
16	297	1.0093	simplex	-0.34998596	6.2772	3.88	3.8703
17	1	0.0000	complex	0.91629073	6.3950	3.16	2.3908
18	132	0.0000	simplex	-0.07006756	6.3422	5.44	2.2952
19	49	0.4719	complex	0.00000000	6.2664	4.68	1.5170
20	103	2.2641	simplex	0.40546511	6.3290	3.20	2.5589
21	0	0.0000	simplex	4.44265126	6.4177	4.00	1.8218
22	6	0.0000	simplex	0.69314718	6.4871	2.68	3.6695
23	7	0.0000	simplex	0.55961579	6.5239	2.00	3.8085
24	85	1.4565	simplex	0.19004360	6.3514	3.40	3.0071
25	29	0.0000	simplex	-0.22314355	6.4089	3.08	3.5957
26	99	0.0594	simplex	-0.77652879	6.3500	5.48	1.8449
27	65	0.0000	simplex	-0.60613580	6.4433	3.36	3.8372
28	30	0.3463	simplex	0.84267891	6.5318	2.52	3.6530
29	53	0.5763	simplex	0.31508105	6.3401	5.04	3.4381
30	142	1.1273	simplex	-1.32525847	6.3036	4.68	1.5349
31	100	1.2335	simplex	0.56666303	6.3451	3.24	3.0987
32	433	0.0000	simplex	-0.74032574	6.3833	5.80	2.8516
33	94	1.0456	simplex	-0.23638878	6.3335	3.76	3.1746

34	19	0.0000	complex	-0.22314355	6.4468	3.88	1.8705
35	19	0.0000	simplex	0.00000000	6.4939	3.68	1.8660
36	184	0.4497	simplex	-0.22992324	6.3261	4.44	3.7255
37	30	0.0000	simplex	-0.21511138	6.3944	3.88	3.5980
38	9	0.0000	complex	1.50407740	6.4022	4.40	1.8716
39	7	0.0000	simplex	2.67414865	6.3753	5.12	2.0139
40	359	1.5565	simplex	0.00000000	6.2984	5.20	3.7201
41	31	0.0000	simplex	0.54159728	6.4393	4.48	1.9295
42	3	0.0000	simplex	0.22314355	6.5161	3.16	2.2484
43	59	1.7296	simplex	-0.91629073	6.4384	2.60	2.7170
44	460	1.6313	simplex	-0.36195692	6.2783	4.48	3.4075
45	303	0.1012	simplex	-0.72659511	6.3860	2.72	4.4303
46	111	0.0277	simplex	0.79088947	6.3179	3.76	3.9283
47	48	0.0000	simplex	-0.80077784	6.6213	2.28	3.6219
48	161	0.8809	simplex	0.44183275	6.3113	3.52	3.5577
49	92	0.0370	simplex	0.56115071	6.3328	3.48	3.9986
50	0	0.0000	simplex	2.83321334	6.4157	4.96	1.9817
51	19	0.0000	simplex	0.30010459	6.3637	4.40	1.7597
52	22	0.0000	simplex	-1.34373475	6.4200	3.68	1.8898
53	22	0.0000	simplex	-0.30228087	6.4277	3.12	1.6785
54	77	0.0000	simplex	-0.03922071	6.3178	5.68	2.0802
55	292	0.2340	simplex	-0.64973469	6.3014	4.04	4.2029
56	316	0.2130	simplex	-1.02270333	6.3220	3.88	4.1927
57	16	0.0000	simplex	0.25782911	6.5924	3.32	3.5920
58	37	0.7235	simplex	1.22994829	6.3692	2.88	3.1948
59	231	0.4960	simplex	0.30900484	6.2452	5.84	2.4730
60	9	0.0000	simplex	0.18232156	6.3173	5.28	1.8289
61	64	0.4621	simplex	-0.39019764	6.5057	3.36	3.6045
62	456	0.6845	simplex	0.48196680	6.2675	5.64	4.3152
63	0	0.0000	simplex	2.48490665	6.4825	3.44	2.0858
64	140	0.5620	simplex	-0.70026465	6.3231	3.60	3.6275
65	8	0.0000	simplex	0.79850770	6.4092	3.40	3.6150
66	14	0.0000	simplex	0.92953596	6.4199	2.20	3.2347
67	854	1.0925	simplex	0.57470603	6.3364	4.84	4.8195
68	137	0.4777	simplex	-0.41639418	6.3365	4.80	1.7232
69	6	0.0000	complex	0.53899650	6.5041	2.04	3.8417
70	7	0.0000	simplex	0.00000000	6.3349	2.48	3.8893
71	93	1.0996	simplex	1.01353328	6.2850	5.60	1.9453
72	6	0.0000	simplex	1.27296568	6.4032	4.12	1.7737
73	200	0.0798	simplex	0.97896184	6.3028	5.68	4.3832
74	95	0.0000	complex	-0.07571182	6.4062	4.08	3.8757
75	78	0.5684	simplex	0.71195651	6.2950	4.12	1.6987
76	46	0.7769	simplex	0.63848877	6.3781	3.00	3.2612

77	19	0.0000	simplex	1.26694760	6.4226	3.92	1.8327
78	23	0.5392	simplex	0.48550782	6.3712	3.56	1.5924
79	162	0.0000	simplex	0.00000000	6.3075	4.00	3.9559
	meanWeight	BNCw	BNCc	BNCd	BNCcRatio	BNCdRatio	
1	3.1806	12.057065	0.000000	6.175602	0.000000	0.512198	
2	2.6112	5.738806	4.062251	2.850278	0.707856	0.496667	
3	1.2081	5.716520	3.249801	12.588727	0.568493	2.202166	
4	1.6114	2.050370	1.462410	7.363218	0.713242	3.591166	
5	4.5167	74.838494	50.859385	241.561040	0.679589	3.227765	
6	1.1365	1.270338	0.162490	1.187616	0.127911	0.934882	
7	1.4951	2.919548	0.649960	18.526805	0.222624	6.345778	
8	3.4178	2.406956	0.812450	1.425139	0.337543	0.592092	
9	0.8244	1.961223	0.000000	0.000000	0.000000	0.000000	
10	1.4860	3.075555	0.162490	1.425139	0.052833	0.463376	
11	3.2365	5.872526	1.949881	1.900185	0.332034	0.323572	
12	3.4517	1.537777	0.974940	0.950093	0.633993	0.617835	
13	1.6507	0.122576	0.000000	0.237523	0.000000	1.937755	
14	1.4315	11.834199	3.087311	22.089653	0.260880	1.866595	
15	3.3756	5.281931	1.624900	5.938079	0.307634	1.124225	
16	3.6470	6.362832	0.649960	4.037893	0.102150	0.634606	
17	1.9674	0.022287	0.000000	0.000000	0.000000	0.000000	
18	1.9716	7.120577	0.487470	20.426991	0.068459	2.868727	
19	1.0740	1.214621	0.000000	5.938079	0.000000	4.888832	
20	2.1666	9.427243	5.199682	6.888171	0.551559	0.730667	
21	1.4082	4.524185	4.549721	6.175602	1.005644	1.365020	
22	3.3214	1.092045	0.324980	1.187616	0.297589	1.087516	
23	3.4878	0.724315	0.000000	0.475046	0.000000	0.655856	
24	2.6612	4.401609	1.949881	1.900185	0.442993	0.431702	
25	3.2684	1.649210	0.162490	1.187616	0.098526	0.720112	
26	1.4437	3.398711	2.437351	14.963958	0.717140	4.402834	
27	3.5655	2.574105	2.112371	2.850278	0.820623	1.107289	
28	3.3654	1.782930	0.487470	1.425139	0.273410	0.799324	
29	3.1251	6.017389	1.137430	23.039745	0.189024	3.828861	
30	1.1317	2.975265	0.649960	1.662662	0.218455	0.558828	
31	2.7558	10.028982	83.194905	5.938079	8.295449	0.592092	
32	2.6131	7.934039	8.286993	25.890023	1.044486	3.263158	
33	2.8385	5.059064	1.462410	10.451018	0.289067	2.065801	
34	1.4442	0.635169	0.162490	0.475046	0.255822	0.747906	
35	1.4424	1.281481	0.000000	0.000000	0.000000	0.000000	
36	3.4551	18.297321	18.848846	10.688542	1.030142	0.584159	
37	3.2720	8.067759	0.324980	0.712569	0.040281	0.088323	
38	1.4581	2.596392	0.649960	6.650648	0.250332	2.561496	
39	1.6395	3.721867	0.974940	7.600741	0.261949	2.042185	

40	3.4881	29.752647	43.872313	22.327176	1.474568	0.750427
41	1.5357	2.117230	0.649960	4.037893	0.306986	1.907159
42	1.8185	0.111433	0.162490	0.237523	1.458184	2.131531
43	2.3265	1.894363	0.324980	0.237523	0.171551	0.125384
44	3.1200	12.636518	10.886833	17.576713	0.861537	1.390946
45	4.2654	3.320707	0.812450	3.087801	0.244662	0.929862
46	3.6724	3.231561	0.324980	3.800370	0.100564	1.176017
47	3.3054	1.270338	0.324980	0.237523	0.255822	0.186976
48	3.2734	7.109434	2.599841	4.275417	0.365689	0.601372
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50	1.5459	1.170048	0.649960	5.938079	0.555499	5.075073
51	1.3216	2.340096	0.649960	3.800370	0.277749	1.624023
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53	1.2208	1.482061	0.000000	0.237523	0.000000	0.160265
54	1.7189	4.524185	1.299920	23.039745	0.287327	5.092573
55	4.0036	12.502798	4.874701	3.800370	0.389889	0.303962
56	3.9911	8.691785	8.611973	7.363218	0.990818	0.847147
57	3.2545	2.718969	0.649960	3.325324	0.239047	1.223009
58	2.8461	12.246502	1.299920	1.900185	0.106146	0.155161
59	2.1948	24.548720	7.799522	40.853981	0.317716	1.664200
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61	3.3145	2.551819	1.462410	0.712569	0.573086	0.279240
62	4.1296	36.940084	18.523866	110.923309	0.501457	3.002790
63	1.6507	0.456876	0.000000	0.475046	0.000000	1.039771
64	3.3489	4.591045	0.162490	3.800370	0.035393	0.827779
65	3.2710	3.276134	0.162490	10.451018	0.049598	3.190046
66	2.8413	1.482061	0.000000	0.000000	0.000000	0.000000
67	4.7138	79.173243	62.071199	59.855833	0.783992	0.756011
68	1.3199	3.264991	0.649960	6.413125	0.199070	1.964209
69	3.5200	1.025185	0.487470	0.000000	0.475495	0.000000
70	3.5673	0.668599	0.000000	2.375231	0.000000	3.552551
71	1.5999	16.681540	6.337112	56.530509	0.379888	3.388806
72	1.3256	0.936038	0.812450	0.950093	0.867967	1.015015
73	4.1964	17.651009	7.312052	100.709814	0.414257	5.705612
74	3.6141	6.719418	0.649960	5.463032	0.096729	0.813022
75	1.2952	3.877873	0.000000	2.850278	0.000000	0.735011
76	2.9268	5.281931	2.599841	2.850278	0.492214	0.539628
77	1.4220	3.454427	0.649960	4.987986	0.188153	1.443940
78	1.1586	2.150660	0.162490	1.187616	0.075554	0.552210
79	3.7049	5.348791	2.924821	14.963958	0.546819	2.797634

```

animals <- data[data$Class == "animal", ]

cherry_rt <- data[data$Word == "cherry", "RT"]

random_thing <- c(subj001[subj001$Word == "cherry", "RT"], data[data$Subject == "J" & data$W

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Advanced

```

data <- lexdec
data$AdjustedRT <- (data$RT / data$Length)
print(data$AdjustedRT)

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 [736] 1.0794953 1.2564534 0.8172675 2.0807410 1.0467327 0.7235272 1.2619836
 [743] 1.1124268 2.1267077 1.5783870 1.2515336 0.8389675 2.1122753 1.3084944
 [750] 1.8609160 0.9254587 0.7423172 0.9143191 1.3297970 1.1441413 0.8462053
 [757] 2.0909590 1.6180865 1.0393950 1.2597898 1.2888262 1.2594218 1.0364333
 [764] 1.0429447 1.2187140 1.2680718 1.3107866 1.7156895 0.9362761 1.2843244
 [771] 1.0334182 0.7089026 1.3359198 1.2780482 0.6445720 0.6415097 0.7903206
 [778] 1.6301553 1.2793860 1.0647602 2.0928403 1.5733548 0.9374920 0.9378950
 [785] 1.1855237 1.0104671 1.0718885 0.6842683 0.6984442 1.6130122 1.4011578
 [792] 0.7115082 2.1553817 1.6106350 1.2797190 1.2409116 0.7036978 0.8322514

[799]	0.9114461	1.2670108	0.7878274	1.2305466	2.0865270	1.0416625	1.5696302
[806]	1.0887223	1.5561395	0.8494633	1.2437200	0.8082681	1.5819843	0.9654421
[813]	2.1295203	1.2935398	0.9159740	0.6347389	1.2739802	1.6560162	2.1453833
[820]	1.5506340	1.2608898	1.0705412	1.3523146	0.9811391	1.0820400	0.8188850
[827]	1.2888262	1.2564534	0.8059133	1.0587715	1.6606218	1.1809835	0.8047190
[834]	1.0432635	1.5815373	0.8958554	0.9037356	0.8467780	2.0928403	0.6276643
[841]	2.1057217	1.5714995	1.3026460	1.5898935	0.7220313	0.8995870	1.0286168
[848]	1.0296573	1.0492110	2.1427017	0.7291604	1.0546562	1.1373957	1.2729502
[855]	1.3874628	1.3014556	0.7342223	1.0805178	0.9055134	2.1063227	0.7955939
[862]	1.2922936	1.0528608	0.9217213	0.6532334	0.8135961	1.0400460	1.2309716
[869]	1.2417180	2.0648017	2.0600057	0.9152686	1.0461072	1.2326630	0.9052609
[876]	1.2515336	1.2396958	1.0222330	2.0600057	0.6161207	0.9969023	0.6864382
[883]	2.0781370	1.5975602	1.0457937	1.5496197	0.8860766	1.0390685	0.8966633
[890]	0.7644865	1.2627096	0.6980297	1.1083798	0.7916600	1.5979793	0.8777409
[897]	0.8920239	0.6902896	1.0857855	1.6256975	0.7862145	0.8010661	1.0489032
[904]	1.5700990	1.2557042	0.8758894	0.9281698	1.0742867	0.9476169	1.7338425
[911]	1.0661550	1.6748750	1.5747373	1.0525597	1.2545754	1.0334182	0.7025072
[918]	1.0438997	1.2464896	2.1227290	0.7714626	2.1874813	2.1116847	1.2457022
[925]	1.2288372	1.2453074	0.7831626	1.2722604	1.3029426	1.2594218	1.2560792
[932]	2.1750100	1.0289643	1.2583138	0.6315358	1.2575718	1.2330836	1.2601572
[939]	1.2182620	0.6135565	1.5686905	0.7878274	1.0817873	0.7765758	0.7750636
[946]	0.6857446	1.5064665	0.8947711	1.2820350	1.0705412	0.6510258	1.2698278
[953]	0.8141538	1.2797190	0.6410175	1.5898935	1.2605238	1.3338996	0.9210437
[960]	0.7354154	2.1175430	0.8176288	1.0479765	0.7987801	0.8006535	0.9080154
[967]	0.9016764	2.1051193	0.7991989	1.0918467	1.6455062	1.1070812	0.7697287
[974]	2.1350760	1.3333914	1.3292782	1.6479185	1.0835483	1.0894487	1.2869094
[981]	1.6253225	0.9067699	1.2922936	1.0942108	1.0694553	1.0587715	1.3064668
[988]	0.8230989	0.7109550	2.0909590	1.0974652	2.1283980	1.6021322	1.1200367
[995]	0.9558346	1.0842973	1.6062173	2.1233003	1.6230600	1.3644394	1.2935398
[1002]	1.2476650	2.1261420	0.8076835	1.6381270	0.9507771	2.1002620	0.9862496
[1009]	0.9436140	1.2894612	0.8222326	0.7985701	1.0661550	1.2803834	1.5950308
[1016]	1.3169582	1.0699992	0.7194926	1.5806412	0.6632002	0.7855167	0.7290034
[1023]	1.0797517	1.2701772	1.2284074	0.9252817	1.2663004	1.3598112	1.2691272
[1030]	0.7831626	2.0157907	1.2605238	0.8366655	1.6733310	0.9764550	1.0549535
[1037]	0.7866774	1.5864090	0.7912151	1.2253738	1.2947782	0.9479924	1.2376528
[1044]	1.0390685	1.0522580	1.5531515	1.3183348	0.7430121	0.7634546	1.3229452
[1051]	0.6582025	1.0889648	1.0820400	0.9143224	0.9133701	1.2944692	1.0251005
[1058]	0.7363020	1.5546500	2.1574487	0.9067699	1.6200112	2.1157963	0.7649430
[1065]	1.0802628	1.0558423	1.2405072	2.1250083	1.0636360	2.0586223	1.3087824
[1072]	1.5450043	1.3138962	1.2663004	1.0098538	1.0251005	1.2712216	1.6461978
[1079]	1.3147360	1.5686905	0.7968781	1.0972348	0.9283260	1.2922936	0.7085252
[1086]	1.6173125	0.8947711	2.0544383	1.2368298	1.6017200	1.0293112	0.7783171
[1093]	0.6276643	0.8080735	1.2279770	1.2392888	0.8325719	0.9210437	1.0655985

[1100]	0.7985701	0.8807740	1.0802628	0.6253829	2.0890670	0.7929850	2.0558387
[1107]	2.0393657	1.2322414	1.2511500	1.0384135	0.7793014	0.7857498	0.6236370
[1114]	0.6142037	1.0374263	1.0549535	1.3326266	1.5894605	0.8256623	1.2836730
[1121]	1.1536097	0.9962204	1.0317192	1.5663253	0.7168943	0.7852834	0.7632238
[1128]	1.0485948	1.3853154	1.3017538	1.3087824	0.9119296	0.7935597	0.9483669
[1135]	1.0540598	1.2872300	1.0641990	0.7191496	1.5868472	0.7357118	1.0397208
[1142]	1.2522984	0.7887397	2.1051193	2.1328650	0.8332105	0.9176067	1.2916676
[1149]	0.9336059	2.1051193	1.2616196	2.0634383	1.2777122	1.5868472	1.0400460
[1156]	0.8897873	1.2339222	1.0531613	1.2284074	0.9516936	1.2750050	1.5273120
[1163]	2.1198580	2.1295203	0.6192362	0.7748099	1.2756852	1.0467327	1.5371170
[1170]	1.0694553	1.5890270	1.0485948	1.5890270	1.5222612	0.7735330	0.7669456
[1177]	0.8608380	1.3164050	1.0200495	1.6000642	1.2437200	0.8037149	2.1063227
[1184]	0.9445773	1.2783834	0.8863654	2.0800920	0.9107179	1.1192305	0.7947302
[1191]	0.7875983	1.0543582	1.2826918	0.6342121	1.5444860	1.2590532	1.3497520
[1198]	0.9544079	1.2553286	0.6450470	0.6593045	1.5470660	1.0661550	0.8134098
[1205]	1.2680718	0.9116880	0.7168943	1.0416625	1.5765688	1.3490472	0.9217213
[1212]	1.0464202	0.6792980	1.2652298	0.8906604	1.2522984	1.2849738	0.9268120
[1219]	1.0787243	2.1311970	2.2339867	1.2813760	1.2507658	1.6406110	1.0006478
[1226]	1.0327407	1.5954540	1.0564323	1.2583138	1.2659442	0.7068113	2.0708687
[1233]	2.1233003	0.8031086	1.6492865	1.2384724	1.6126175	1.0540598	0.7138743
[1240]	0.7896456	2.0281663	2.1087163	0.9406844	1.0558423	0.8120969	1.5971403
[1247]	1.0448493	1.2368298	1.0313773	1.2401018	1.2619836	1.5971403	0.9310959
[1254]	0.7120588	1.2146090	1.5797420	0.8070962	0.7916600	1.0300028	0.8878011
[1261]	1.2575718	1.0170932	2.0978063	0.7717232	0.9010750	0.7997982	1.3693886
[1268]	1.1382990	1.1679335	1.1370328	1.3527770	0.8714586	1.0839016	0.7579008
[1275]	1.4075812	0.6835185	1.3261366	1.1484348	2.2008627	1.6812585	0.8771883
[1282]	1.6434200	1.1616318	0.8790780	1.3696010	1.1342872	1.7415060	1.1359400
[1289]	1.3462036	1.3986030	1.1280822	2.1589907	0.7711358	1.2743224	2.1754983
[1296]	1.7599150	0.9784380	0.6884487	1.3629086	1.0021163	0.8592747	1.0320036
[1303]	1.3773064	1.6921233	1.6420195	0.8625914	1.0167437	0.7055313	1.1194325
[1310]	1.6475753	0.8317691	1.3133344	2.1645847	1.3023490	0.9687226	0.7568049
[1317]	1.1462160	0.8516806	0.9598590	1.1257673	1.3591412	1.4107172	1.6455062
[1324]	1.7030862	1.1453547	2.2093470	0.9331906	1.6327195	1.1006573	1.3138962
[1331]	2.1944697	1.3536986	0.9719293	1.4072296	0.8246433	2.1715710	0.7470924
[1338]	2.1808240	0.9029120	1.3005580	1.1755977	1.1122163	1.3771020	1.3843316
[1345]	1.3916896	2.1421627	1.3513864	1.7112037	2.1238707	1.2987508	1.6509860
[1352]	1.2677188	1.2894612	1.0857855	1.6226810	1.1255730	0.8413772	1.6420195
[1359]	1.2813760	1.6506470	1.5992325	0.9687226	0.7523099	1.2904098	0.8819141
[1366]	0.9199071	1.2756852	0.9644430	1.2984480	0.8256623	0.8491832	2.1157963
[1373]	1.0899310	2.1075217	1.0734912	1.0983835	0.9289700	1.0875050	1.3406376
[1380]	1.6110327	0.7251661	1.0587715	0.8701620	0.7477582	2.1389107	1.6391945
[1387]	1.0970042	0.9596864	1.3691760	1.3253436	1.6234385	1.1218363	1.3766926
[1394]	1.0625042	1.3175100	1.2852976	1.6271923	2.2656080	1.1326177	0.7145923

```

[1401] 0.8520920 0.8215354 1.2708740 1.2756852 0.9325654 0.6561031 0.9906281
[1408] 0.8134098 1.3005580 0.6788972 0.9230669 0.8529108 0.7488152 0.6679599
[1415] 1.0710813 2.1300803 1.0792388 1.0708115 2.1272720 1.1031232 0.9221713
[1422] 0.8335285 1.2966214 1.0549535 2.1272720 0.7058477 0.8969316 2.0728667
[1429] 0.7797906 0.8213604 1.5868472 0.8869414 1.0406945 1.2355888 0.6202536
[1436] 1.0633538 0.7682911 1.0229545 0.8201289 1.5954540 0.8158119 0.6366470
[1443] 1.0573133 1.2579432 2.1099070 1.2305466 1.5450043 0.7923242 0.6146329
[1450] 1.0423048 0.7727606 1.2258100 0.8892226 1.5360465 1.2218496 1.2372418
[1457] 2.0675120 0.9214957 1.5506340 2.0928403 0.9032236 1.2484446 1.5153642
[1464] 1.0204155 1.0300028 1.5349712 0.8082681 1.2090010 2.1700860 0.8974667
[1471] 1.2627096 1.2396958 0.8895053 1.0800075 0.8903497 1.2634330 0.6935803
[1478] 0.6907340 2.0408310 1.5677470 1.2284074 1.0354343 1.2376528 1.5541515
[1485] 1.5376508 0.7812469 1.2449116 2.0668363 1.0390685 1.2266796 1.2619836
[1492] 0.9082631 1.2608898 1.0380852 1.0448493 0.6942270 1.2904098 0.6935803
[1499] 0.7838735 1.0570202 1.0330798 0.9464846 0.7251661 0.8020915 0.8002396
[1506] 0.6444131 1.0561377 2.0648017 1.0797517 1.1650428 0.6579251 2.1665957
[1513] 1.2790524 1.7005987 2.1754983 1.3213300 0.8293291 1.6423702 1.0830470
[1520] 2.1453833 2.1774447 0.9192210 0.9136089 0.9416677 0.8222326 1.3067578
[1527] 1.6399037 0.9700139 1.1359400 0.7587458 1.2803834 0.6693324 1.2708740
[1534] 1.6268195 1.3055916 1.3359198 1.7523522 2.1002620 1.2862662 0.7493399
[1541] 1.3067578 1.6058117 1.2670108 0.8303109 1.2990532 1.0683625 1.0317192
[1548] 2.0965720 1.5397737 0.7058477 0.7845804 1.0390685 1.0531613 1.0570202
[1555] 1.2594218 1.5907570 0.7050707 1.2637936 1.2515336 1.5691607 1.2760246
[1562] 0.7829247 1.0528608 1.2763632 0.8316079 1.3548448 0.9418636 1.0726917
[1569] 1.0454795 1.2413152 1.0236728 0.9586461 0.9114461 0.8128488 0.8834499
[1576] 0.8033110 2.0735300 1.5491110 1.2926058 1.0561377 1.1070812 1.5801920
[1583] 1.0678133 1.2926058 1.2780482 0.9445773 1.0981543 0.8487621 0.9504090
[1590] 2.1204340 1.4223024 0.8397256 1.3425912 1.0750783 1.3213300 2.1361763
[1597] 0.7403481 1.3017538 1.2719148 1.0587715 0.8806131 2.1272720 0.9217213
[1604] 0.9542284 1.1120055 1.6742585 0.8567944 1.0699992 1.2691272 1.1314953
[1611] 0.9199071 1.3052990 2.0978063 0.6905753 2.2495867 1.3017538 0.8374375
[1618] 1.3099302 1.5937562 0.8787466 0.8553354 1.6944117 0.7625287 1.0951472
[1625] 1.1959150 1.7500835 1.3023490 2.1069227 1.1171980 0.9720879 0.7501223
[1632] 0.9457247 1.0932692 0.7920551 1.2900940 0.9863930 1.2677188 1.2929176
[1639] 1.5792913 1.0884797 1.6374127 0.7529452 1.5971403 1.3331368 0.6588926
[1646] 1.2878700 1.0540598 0.6493754 0.8464920 1.0522580 1.3055916 2.2356207
[1653] 2.0846097 0.8325719 1.6264460 0.8451966 0.9688846 1.5546500 1.2823636

```

```
tapply(data$RT, data$Class, mean)
```

```

  animal    plant
6.387746 6.381751

```



```
tapply(data$RT, data$Subject, mean)
```

A1	A2	A3	C	D	I	J	K
6.277734	6.220381	6.397734	6.322315	6.406071	6.253121	6.254676	6.192831
M1	M2	P	R1	R2	R3	S	T1
6.173212	6.532709	6.429181	6.307263	6.472234	6.392904	6.373671	6.362803
T2	V	W1	W2	Z			
6.807491	6.587316	6.263898	6.460466	6.598874			

```
tapply(data$RT, data$Word, mean)
```

almond	ant	apple	apricot	asparagus	avocado	banana
6.432969	6.323140	6.245157	6.363695	6.481501	6.403154	6.317822
bat	beaver	bee	beetroot	blackberry	blueberry	broccoli
6.307590	6.419903	6.297762	6.493389	6.395947	6.418705	6.408560
bunny	butterfly	camel	carrot	cat	cherry	chicken
6.409159	6.406196	6.385989	6.350004	6.267518	6.342565	6.283576
clove	crocodile	cucumber	dog	dolphin	donkey	eagle
6.427672	6.464146	6.439320	6.295568	6.369348	6.321655	6.369188
eggplant	elephant	fox	frog	gherkin	goat	goose
6.380212	6.395361	6.332288	6.333516	6.557580	6.274979	6.338759
grape	gull	hedgehog	horse	kiwi	leek	lemon
6.303557	6.460136	6.568672	6.336376	6.317260	6.456525	6.269145
lettuce	lion	magpie	melon	mole	monkey	moose
6.375319	6.301367	6.552365	6.371231	6.414975	6.307494	6.366872
mouse	mushroom	mustard	olive	orange	owl	paprika
6.319791	6.336505	6.417676	6.294963	6.284955	6.358219	6.505205
peanut	pear	pig	pineapple	potato	radish	reindeer
6.299883	6.335290	6.256939	6.402199	6.360034	6.447520	6.529878
shark	sheep	snake	spider	squid	squirrel	stork
6.317934	6.298444	6.339244	6.340053	6.408912	6.490344	6.563291
strawberry	swan	tomato	tortoise	vulture	walnut	wasp
6.347469	6.378116	6.342152	6.563449	6.649979	6.436142	6.529231
whale	woodpecker					
6.354825	6.458256					

Super Mega Giga Insane Extreme Relentless Task

```
participants_avg <- tapply(data$RT, list(data$Subject, data$Class), mean)
print(participants_avg)
```

	animal	plant
A1	6.289626	6.262783
A2	6.217799	6.223626
A3	6.388859	6.408891
C	6.323871	6.320358
D	6.418244	6.390767
I	6.281769	6.217107
J	6.280086	6.222733
K	6.181169	6.207491
M1	6.171069	6.175906
M2	6.549926	6.511064
P	6.453780	6.398257
R1	6.301740	6.314206
R2	6.451372	6.498460
R3	6.382524	6.405954
S	6.384995	6.359434
T1	6.369048	6.354953
T2	6.809137	6.805422
V	6.595919	6.576501
W1	6.242581	6.290697
W2	6.441990	6.483693
Z	6.607151	6.588468

```
t.test(participants_avg[, "plant"], participants_avg[, "animal"], paired = T)
```

Paired t-test

```
data: participants_avg[, "plant"] and participants_avg[, "animal"]
t = -0.81872, df = 20, p-value = 0.4226
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -0.021267781  0.009278577
sample estimates:
mean difference
 -0.005994602
```

Loops and Conditionals

Basic + Intermediate + Advanced

```
thingamadob <- 68
if (thingamadob %% 2 == 0) {
  print("Hello, World!")
} else {
  print(" ")
}
```

```
[1] "Hello, World!"
```

```
for (i in 1:20) {
  print(i)
}
```

```
[1] 1
[1] 2
[1] 3
[1] 4
[1] 5
[1] 6
[1] 7
[1] 8
[1] 9
[1] 10
[1] 11
[1] 12
[1] 13
[1] 14
[1] 15
[1] 16
[1] 17
[1] 18
[1] 19
[1] 20
```

```

numbers <- c(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23)
)
for (i in numbers) {
  if (i <= 20) {
    print(i)
  } else {
  }
}

```

```

[1] 1
[1] 2
[1] 3
[1] 4
[1] 5
[1] 6
[1] 7
[1] 8
[1] 9
[1] 10
[1] 11
[1] 12
[1] 13
[1] 14
[1] 15
[1] 16
[1] 17
[1] 18
[1] 19
[1] 20

```

```

n <- 12
pi <- 0
for (i in 1:100) { # Change 100 to n for manual setting
  if( i %% 2 == 1) {
    pi <- pi + 1 / (2*i - 1)
  } else {
    pi <- pi - 1 / (2*i - 1)
  }
  print(pi * 4)
}

```

```

[1] 4

```

[1] 2.666667
[1] 3.466667
[1] 2.895238
[1] 3.339683
[1] 2.976046
[1] 3.283738
[1] 3.017072
[1] 3.252366
[1] 3.04184
[1] 3.232316
[1] 3.058403
[1] 3.218403
[1] 3.070255
[1] 3.208186
[1] 3.079153
[1] 3.200366
[1] 3.08608
[1] 3.194188
[1] 3.091624
[1] 3.189185
[1] 3.096162
[1] 3.18505
[1] 3.099944
[1] 3.181577
[1] 3.103145
[1] 3.178617
[1] 3.10589
[1] 3.176065
[1] 3.108269
[1] 3.173842
[1] 3.11035
[1] 3.171889
[1] 3.112187
[1] 3.170158
[1] 3.11382
[1] 3.168615
[1] 3.115281
[1] 3.167229
[1] 3.116597
[1] 3.165979
[1] 3.117787
[1] 3.164845
[1] 3.118868

[1] 3.163812
[1] 3.119856
[1] 3.162867
[1] 3.120762
[1] 3.161999
[1] 3.121595
[1] 3.161199
[1] 3.122364
[1] 3.160459
[1] 3.123076
[1] 3.159773
[1] 3.123737
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[1] 3.124353
[1] 3.158541
[1] 3.124927
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[1] 3.125465
[1] 3.157465
[1] 3.125969
[1] 3.156976
[1] 3.126442
[1] 3.156517
[1] 3.126888
[1] 3.156085
[1] 3.127308
[1] 3.155676
[1] 3.127704
[1] 3.155291
[1] 3.12808
[1] 3.154925
[1] 3.128435
[1] 3.154579
[1] 3.128773
[1] 3.15425
[1] 3.129093
[1] 3.153938
[1] 3.129398
[1] 3.15364
[1] 3.129688
[1] 3.153357
[1] 3.129965
[1] 3.153087

```
[1] 3.130229
[1] 3.152828
[1] 3.130482
[1] 3.152581
[1] 3.130723
[1] 3.152345
[1] 3.130955
[1] 3.152119
[1] 3.131176
[1] 3.151902
[1] 3.131389
[1] 3.151693
[1] 3.131593
```

Super Advanced

```
week <- c("monday", "tuesday", "wednesday", "thursday", "friday")
daytime <- c("8am", "9am", "10am", "11am", "12pm", "1pm", "2pm", "3pm", "4pm", "5pm", "6pm")
btn_mon_fri <- 0.40
aft_mon_fri <- 0.25
sat_sun <- 0.15
day <- "tuesday"
time <- "4pm"
length_of_call <- 20

if (day %in% week & time %in% daytime) {
  cost <- (length_of_call * btn_mon_fri)
} else {
  if (day %in% week & !(time %in% daytime)) {
    cost <- (length_of_call * aft_mon_fri)
  } else {
    if (!(day %in% week)) {
      cost <- (length_of_call * sat_sun)
    }
  }
}
print(cost)
```

```
[1] 8
```